Project Milestones

Milestones M1–M9 and their completion criteria

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Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 20 October 2026

1 Purpose

This document lists the nine milestones of the LipScribe implementation, when each is due, and the criteria used to decide that it is complete. The milestones come from Section 12 of the *Technical Implementation and System Development Document* and are tracked on the team's Trello board, where the Work Breakdown Structure (WBS) breaks each milestone into cards with an owner, a reviewer and a due date.

The implementation runs for six weeks, from Monday 19 October to Sunday 29 November 2026. Each week ends with a milestone review on Sunday, chaired by the project coordinator (Charis Opol), where the completion criteria are checked and the next week is re-planned.

2 Timeline

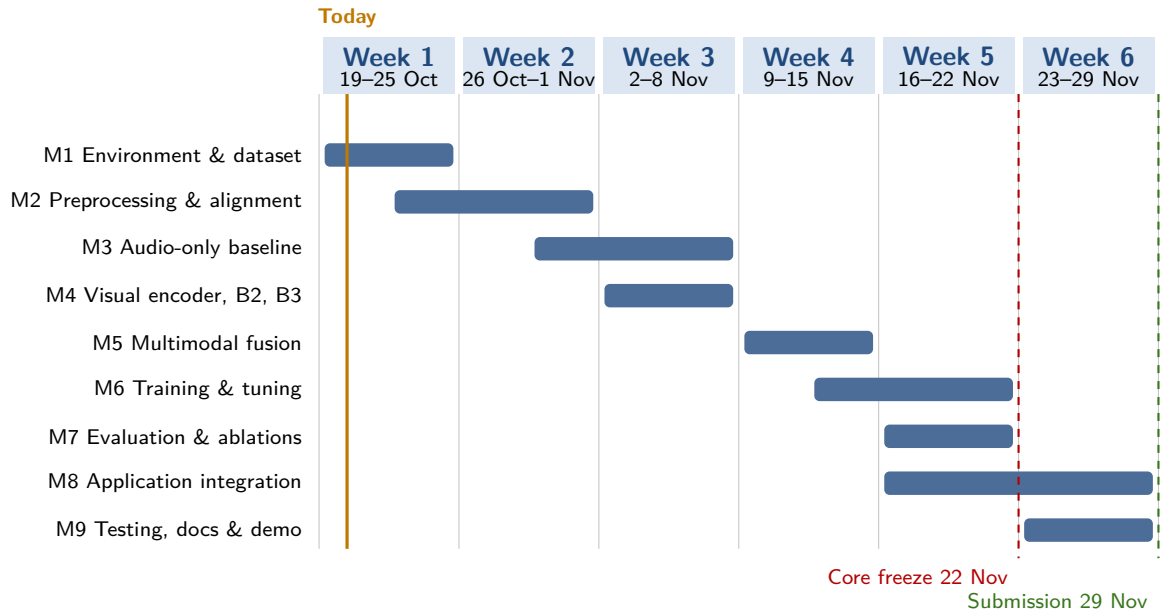

Figure 1: Milestone schedule over the six implementation weeks. Bars show the weeks in which each milestone's work takes place; the review is at the end of the last week of each bar.

3 Milestones and completion criteria

Milestone	Review	Completion criteria	Status
M1 Environment setup and dataset verification	Week 1 Sun 25 Oct	Every member runs the test suite; GRID integrity report lists usable utterances per speaker; speaker-independent `splits.csv` committed and disjoint; GPU access confirmed; mouth-crop prototype approved.	In progress
M2 Preprocessing and data alignment	Week 2 Sun 1 Nov	Mouth crops cached and contact sheets reviewed; noise-mixing tests pass; one batch loads with consistent lengths; train/inference parity test passes; audio-only baseline (B1) full run started.	Planned
M3 Audio-only baseline	Week 3 Sun 8 Nov	B1 evaluated at all five noise levels (clean, 10, 5, 0, -5 dB); `evaluate.py` and beam search merged.	Planned
M4 Visual encoder, B2 and B3	Week 3 Sun 8 Nov	Encoder shape tests pass; visual-only B2 validation CER clearly below an untrained model; concatenation baseline B3 evaluated at all noise levels.	Planned
M5 Multimodal fusion	Week 4 Sun 15 Nov	Fusion unit tests pass with no NaN under modality dropout; proposed model P trains stably and has first results; ablation configs X1–X3 ready; `transcribe.py` merged.	Planned
M6 Training and tuning	Week 5 Sun 22 Nov	Learning-rate and modality-dropout sweeps logged; final configuration chosen on validation data only; final runs of B1, B2, B3 and P complete.	Planned
M7 Evaluation and ablations	Week 5 Sun 22 Nov	All core models evaluated on identical test audio; ablations X1–X3 run; every number traceable to a config and checkpoint; repository tagged `v1.0-core` (core freeze).	Planned
M8 Application integration	Weeks 5–6 Tue 24 Nov	Gradio upload demo (Mode A) and live captioning (Mode B) working; latency measured (target real-time factor ≤ 0.5); demo captions match `evaluate.py` on held-out clips.	Planned
M9 Testing, documentation and demonstration	Week 6 Sun 29 Nov	Fresh clone installs and reproduces one evaluation; demo rehearsed with a recorded fallback video; all automated tests pass; report, code and release (`v1.0-final`) submitted.	Planned

4 Current status: Milestone M1

Implementation started on Monday 19 October 2026 according to the WBS. Progress against the M1 criteria as of 20 October:

- ☒ Trello board set up with weekly lists, owners, reviewers and due dates
- ☒ GitHub repository created
- ☒ GRID dataset downloaded
- ☒ GPU access secured for training
- ☐ Pinned environment and test suite running for all six members
- ☐ Integrity report and speaker-independent split (`splits.csv`)

☐ Mouth-crop prototype approved

The details are given in the accompanying Progress Report (Week 1).

5 Review process and fallback plan

- A milestone is complete only when all its criteria are met and the reviewer of each card has approved the work. Unfinished cards move to the next week with a written reason.
- If the schedule slips, work is dropped in this order: optional extensions; experiments X4–X6; live Mode B (Mode A is kept); extra random seeds.
- Never dropped: baselines B1–B3, the proposed model P, the per-noise-level evaluation and ablations X1–X3.